

2025 Invitational Tournament - Div C -Georgia

Designer Genes - MCQ TEST PACKET



Written by: Pragnya Kamalapuri, Yong Zhu, Shreejanani
Thirugnanasambantham, Leah Green

School Name/Team: _____

Student Names: _____

Instructions:

1. Write all answers on the Answer Sheet. DO NOT WRITE on the Test Packet.
2. See the Image Sheet when asked to refer to an image by a question (it may be a separate sheet or attached to the back of this packet).
3. For numerical questions, answers within a reasonable range will be accepted.
4. When answering free-response questions, use the appropriate abbreviations for satellite and instrument names
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Mendelian Genetics

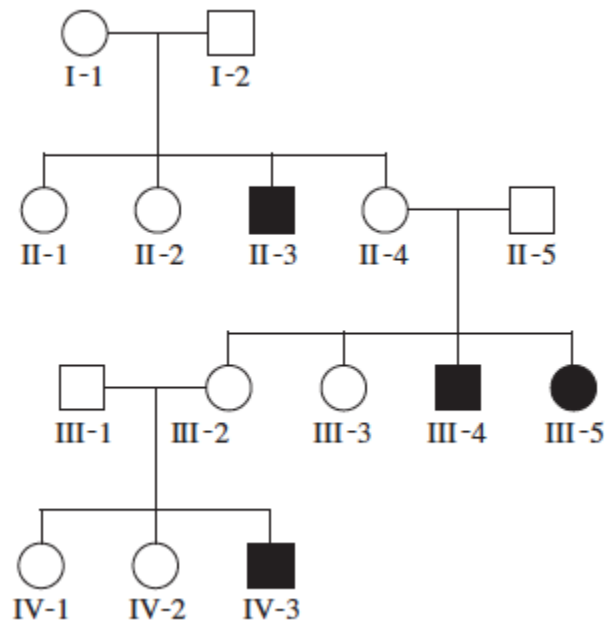
1. Alex and Maria are considering having children. Both are unaffected but each has a younger sister affected with a very rare genetic disorder. Their parents are unaffected. If the trait is autosomal recessive with complete penetrance, what is the probability that Alex and Maria's first child will be affected? [1]
 - a. 1/4
 - b. 1/9
 - c. 2/9
 - d. 1/16

2. Genes that control coat color in mammals often exhibit multiple alleles. In hamsters, the C gene has three alleles: C (black fur), c^k (sepia fur), and c^a (albino fur). Each allele supplies a different amount of pigment. If geneticists cross a $c^a c^a$ albino hamster with a $c^k c^k$ sepia hamster, what outcome would support the conclusion that c^a and c^k are alleles of the same gene, rather than mutations in two different coat color genes? [1]
 - a. All offspring have black fur
 - b. All offspring have sepia fur
 - c. All offspring have albino fur
 - d. Offspring show both albino and sepia phenotypes in a 1:1 ratio

3. Polydactyly is a condition in which a person has extra fingers and/or toes. It is caused by a dominant allele. If a person carries this allele but does not have any extra fingers or toes, this is an example of? [1]
 - a. haploinsufficiency
 - b. a dominant negative mutation
 - c. incomplete penetrance
 - d. a gain-of-function mutation

4. Which of the following phenotypes is not an example of a wildtype phenotype? [1]
 - a. Yellow-flowered elderflower orchid
 - b. Red-flowered elderflower orchid
 - c. A gray elephant
 - d. An albino (white) elephant

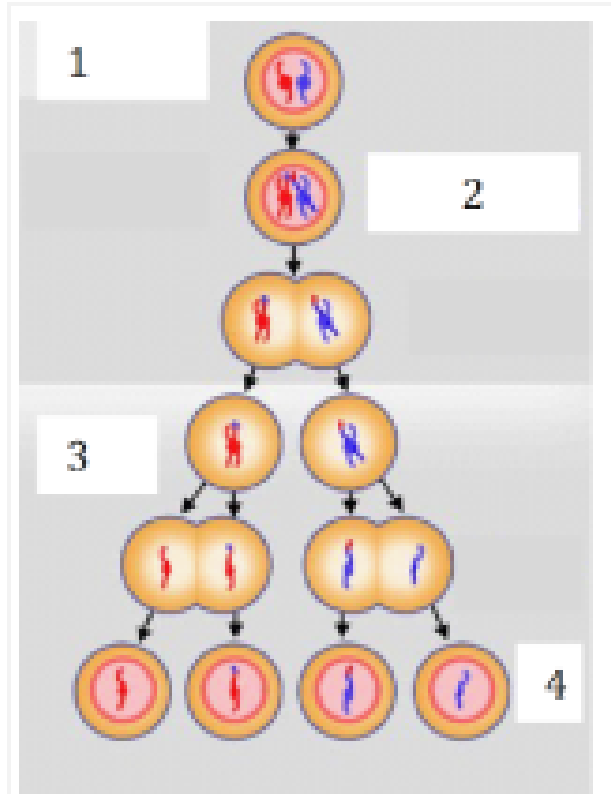
5. A cross is made between a white-eyed female fruit fly and a red-eyed male. What would be the reciprocal cross? Assume W is the allele for white eyes and W^+ is the allele for red eyes. [*1]
- Female is $X^W X^W$ and male is $X^W Y$.
 - Female is $X^{W^+} X^{W^+}$ and male is $X^{W^+} Y$.
 - Female is $X^{W^+} X^{W^+}$ and male is $X^W Y$.
 - Female is $X^W X^W$ and male is $X^{W^+} Y$.
6. Which of the following is a possible explanation for pleiotropy? [1]
- The expression of a single gene can affect cell function in more than one way.
 - A gene may be expressed in different cell types in a multicellular organism.
 - A gene may be expressed at different stages of development.
 - All of the above are possible explanations
7. Propose the most likely mode of inheritance (autosomal dominant, autosomal recessive, or X-linked recessive) for the disease represented in the following pedigree. Affected individuals are shown as filled (black) symbols. [1]



- Autosomal dominant
- Autosomal recessive
- X-linked recessive
- None of the above

Mitosis and Meiosis

8. Which of these mechanisms are ways in which meiosis increases the diversity within a population? [1]
 - a. centromere sliding and telomere inversion
 - b. independent assortment of chromosomal homologs and chromosomal non-disjunction
 - c. independent assortment of chromosomal homologs and homologous non-sister chromatid exchange
 - d. Spontaneous introduction of transposable elements and microsatellite repeat expansion
9. Klinefelter syndrome occurs in approximately 1 in 500 live births. People with this syndrome have XXY chromosomal make up. What causes this to happen? [1]
 - a. Non-disjunction of the X-chromosome in either the mother or father
 - b. Non-disjunction of X-chromosome in the mother OR of the X-Y in the father
 - c. Non-disjunction of X chromosome in the mother
 - d. Non-disjunction of X chromosome in the father
10. At which stage of the mitotic cell cycle is there the highest ratio of DNA to cytoplasm? [*1]
 - a. Beginning of metaphase
 - b. End of G2 phase
 - c. Beginning of G1 [phase
 - d. End of S phase
11. When do chromosomes condense during prophase I of meiosis? [1]
 - a. Diakinesis
 - b. Diplotene
 - c. Zygotene
 - d. Leptotene
12. Fruit flies (*Drosophila melanogaster*) have a diploid number of chromosomes of eight (8). Therefore, their haploid number is four (4). How many different ways can fruit fly homologues be combined? [1]
 - a. 256 different ways
 - b. 16 different ways
 - c. 4 different ways
 - d. 2 different ways



13. Use the diagram above to answer the following question: the area labeled 1 represents which type of cellular reproduction? [1]

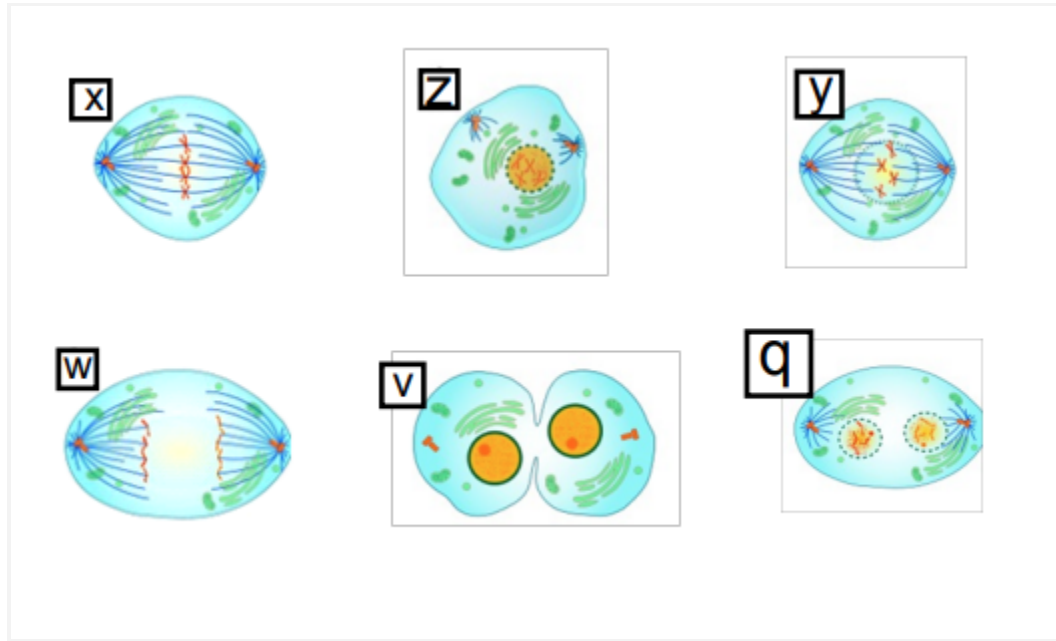
- mitosis 1
- mitosis 2
- meiosis 1
- meiosis 2

14. When treating many types of cancer, the treatments often result in loss of hair (as one side effect). Why is this a common side effect? [1]

- Treatments target cells that divide infrequently, such as cancer cells and hair follicle cells.
- Treatments target cells that divide slowly, such as cancer cells and hair follicle cells.
- Treatments target cells that divide quickly, such as cancer cells and hair follicle cells.
- Treatments are administered and the hair follicle cells do not have enough resources to reproduce

15. Polyploidy is lethal in human beings, however for some organisms it can result in larger organisms. Which organisms may flourish under polyploidy conditions? [1]

- a. Protists
- b. Bacteria
- c. prokaryotic cells
- d. plants



16. Which of the following represents the correct sequence, based upon the images above?

[1] (the letters used to identify each image have no bearing on the image itself)

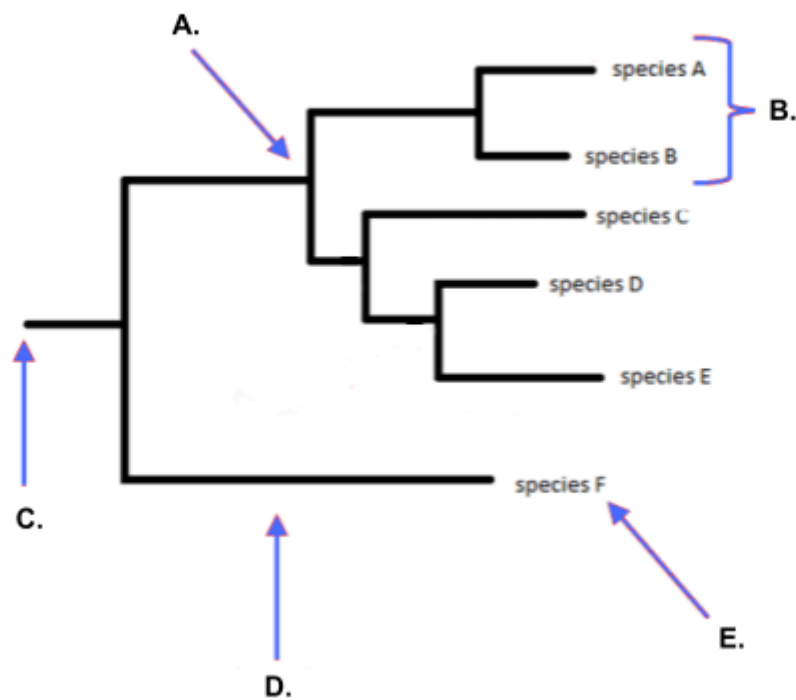
- a. X,z,y,w,v,q
- b. Q,w,x,y,v,z
- c. V,z,y,x,w,q
- d. z,y,x,w,q,v

Population and evolutionary genetics:

The following information is for questions 17-19: A group of research students are doing a field study on a squirrel population concerning tail morphology. They observed that in a population of 500 squirrels, 80 had thin tails and this population is currently in Hardy Weinberg equilibrium. It is also known that bushy tails (T) as the dominant trait and thin tails (t) as a recessive trait.

17. For a population to be in Hardy-Weinberg equilibrium, which of the following must be true? [1]
- a. Population size must be small
 - b. There must be no mutations in the population
 - c. The population partakes in random mating
 - d. The population undergoes natural selection
18. What is the frequency of the dominant allele (T) in this squirrel population as a percent? [*1]
- a. 84%
 - b. 60%
 - c. 53%
 - d. 42%
19. What is the frequency of the heterozygous genotype in this squirrel population as a percent? [*1]
- a. 76%
 - b. 53%
 - c. 48%
 - d. 24%

Figure 1



20. According to Figure 1, what is the relationship between species A and B?[1]
- Orthologs
 - Paralogs
 - Analog
21. According to Figure 1, what is the relationship between species B and C?[1]
- Orthologs
 - Paralogs
 - Analog
22. Which of the following accurately shows the chronological process of constructing a phylogenetic tree? [1]
- Data Acquisition and Preparation, Model Selection, Multiple Sequence Alignment, Tree construction, and Tree Evaluation
 - Model Selection, Data Acquisition and Preparation, Multiple Sequence Alignment, Tree construction, and Tree Evaluation
 - Data Acquisition and Preparation, Multiple Sequence Alignment, Model Selection, Tree construction, and Tree Evaluation
 - Data Acquisition and Preparation, Multiple Sequence Alignment, Model Selection, Tree Evaluation, and Tree Construction
23. Which statistical technique is commonly used after a phylogenetic tree has been constructed to assess the reliability and support for each specific branching point (node)?[*1]
- Multiple Sequence Alignment (MSA).
 - Bootstrapping.
 - Calculation of the chi-squared statistic.
 - Molecular clock analysis.
24. What is the primary goal of the multiple sequence alignment (MSA) step in phylogenetic construction?[1]
- To create the final tree topology based on sequence similarity.
 - To randomly delete sites that contain gaps to simplify the data.
 - To ensure that homologous sites across different sequences are correctly matched.
 - To convert DNA sequences into amino acid sequences.

Molecular biology of DNA

25. Which of the following best describes the characteristic of DNA in Chargaff's Rule?[1]
- a. DNA always contains equal amount of purines and pyrimidines because of specific base pairing of the nitrogenous bases
 - b. The sequence of the nucleotides in DNA determines the trait of an organism
 - c. All organisms contain the sugar-phosphate backbone in the DNA, just at different base ratios.
 - d. DNA has an unique double helix structure that allows it to have a stable structure for DNA replication
26. Which of the following correctly describes a nucleoside?[1]
- a. A nucleotide consists of a nitrogenous base, a phosphate group, and a 5 carbon sugar
 - b. A nucleoside is formed of a nitrogenous base binding to a 5 carbon sugar and can form DNA monomer by bonding with phosphate groups
 - c. A nucleoside and a nucleotide are very similar chemically, except nucleotide has an additional 5 carbon group and one is a monomer of DNA
 - d. A nucleoside contains of only a nitrogenous base and carboxyl group with the lacking of phosphate group
27. In DNA, the phosphate group is attached to which carbon of the deoxyribose sugar?[1]
- a. 1'
 - b. 2'
 - c. 3'
 - d. 5'
28. DNA replication is considered semi-conservative because:[1]
- a. Each daughter molecule contains two entirely new strands.
 - b. Each daughter molecule contains one parental strand and one new strand.
 - c. The parent DNA molecule is conserved, and two new strands are synthesized separately.
 - d. Each strand alternates between old and new nucleotides.
29. Why are purine–purine or pyrimidine–pyrimidine base pairs not used in the DNA?[1]
- a. Their hydrogen bonds would be too strong to allow strand separation.
 - b. They would disrupt the uniform diameter of the double helix.
 - c. Their hydrogen bonding is too weak to stabilize the double helix.
 - d. DNA polymerase is unable to recognize them during replication.

30. A chemotherapy drug interferes with thymine production, leading to increased incorporation of uracil into DNA. What is the most likely cellular consequence of this substitution?[1]
- a. The DNA helix becomes more stable because uracil forms stronger hydrogen bonds.
 - b. DNA repair systems are triggered, leading to strand breaks and cell death.
 - c. Replication proceeds normally since uracil pairs correctly with adenine.
 - d. The sugar-phosphate backbone of DNA is disrupted, causing chromosomes to fragment spontaneously.
31. Which of the following is NOT a function of chromatin?[1]
- a. Compacting DNA to fit inside the nucleus.
 - b. Protecting DNA from damage.
 - c. Producing ribosomal RNA in the nucleolus.
 - d. Helping regulate gene expression.
32. ATP (adenosine triphosphate) is often the source of energy for the cell. Structurally, why is it classified as a nucleotide rather than just a nucleoside?[1]
- a. Because it contains adenine linked only to deoxyribose.
 - b. Because it has a nitrogenous base but no phosphate groups.
 - c. Because it contains a nucleoside along with three phosphate groups attached.
 - d. Because it lacks a sugar component needed to in a nucleoside
33. Why are centromeres and telomeres usually made of heterochromatin?[1]
- a. They contain many protein-coding genes that need constant expression.
 - b. They are specialized regions commonly in ribosomal RNA transcription and nucleolus assembly.
 - c. Their tightly packed structure maintains stability and reduces unwanted recombination.
 - d. Their loose packing allows them to replicate and transcribe more quickly.

Prokaryotic gene expression and regulation

34. Select which choice accurately describes the Central Dogma. [1]
- a. DNA is translated to RNA which is translocated to protein.
 - b. RNA is transcribed to DNA which is translated to protein.
 - c. Protein is translated to DNA which is transcribed to RNA.
 - d. DNA is transcribed to RNA which is translated to protein.

35. Select the correct directionality of transcription for the leading strand. [1]
- DNA is read 3' to 5' and the corresponding RNA strand is built 5' to 3'
 - DNA is read 5' to 3' and the corresponding RNA strand is built 3' to 5'.
 - RNA is read 3' to 5' and the corresponding DNA strand is built 5' to 3'
 - RNA is read 5' to 3' and the corresponding DNA strand is built 3' to 5'
36. During the initiation stage of transcription, which enzyme binds to which site?[1]
- DNA polymerase; start codon
 - RNA polymerase; promoter
 - DNA ligase; operon
 - Primase; primer
37. True or false: When pre-mRNAs undergo splicing, exons are removed from the sequence, leaving the mature mRNA. [1]
- True
 - False
38. Hypothetical gene X is expressed when there is an excess of glucagon in the body, working to return glucagon levels to baseline. The gene X operon could be described as... [1]
- Positive inducible
 - Negative inducible
 - Positive repressible
 - Negative repressible
39. Place the following steps in order of *lac* operon function when lactose is present and glucose levels are low: *[1]

- (1) RNA polymerase is activated.
- (2) cAMP levels increase due to low glucose levels and a halt of ATP production.
- (3) Lactose binds to the *LacI* repressor, causing a conformational change and leading the repressor to fall off the operator.
- (4) RNA polymerase binds to the promoter and transcribes the *lac* gene
- (5) cAMP binds to CAP, then binding to the promoter of the *lac* operon

- a. 3,1,2,5,4
- b. 1,3,5,3,4
- c. 2,5,3,1,4
- d. 1,4,3,2,5

40. What does the “m” in “mRNA” stand for, in regard to translation? [1]
- a. Micro
 - b. Messenger
 - c. Multiple
 - d. Molecular
41. To start any DNA coding sequence, a “start codon” must be present. Which amino acid is the correct “start codon”? [1]
- a. Arginine
 - b. Asparagine
 - c. Tryptophan
 - d. Methionine
42. What is the purpose of the 5’ cap added to mRNA after transcription? [1]
- a. To guide the mRNA to the ribosome
 - b. To tell translation where to start
 - c. To protect the mRNA from degradation
 - d. To give tRNA something to attach onto during translation
43. What is the difference between a cis- and trans-regulatory element? [1]
- a. Cis-regulatory elements are on the same chromosome as the gene they control, and trans-regulatory elements are located on any chromosome and are able to travel.
 - b. Trans-regulatory elements are on the same chromosome as the gene they control, and cis-regulatory elements are located on any chromosome and are able to travel.
 - c. Cis-regulatory elements are on the leading DNA strand and trans-regulatory elements are on the lagging DNA strand.
 - d. Trans-regulatory elements are on the leading DNA strand and cis-regulatory elements are on the lagging DNA strand.

Technology and Techniques

44. Illumina sequencing is a next-generation sequencing method that is able to sequence high volumes of DNA sequences quickly. Name the method in which Illumina is able to do this. [1]
- a. Polymerase-chain reaction (PCR)
 - b. Sequencing-by-synthesis (SBS)
 - c. Large-scale gel-electrophoresis
 - d. Mutation-identification technology

45. What unique ability of plasmids allows them to be critical tools for molecular cloning?

[1]

- a. Plasmids are separate from chromosomal DNA, and are able to replicate independently
- b. Plasmids are strong, circular fragments of DNA that are unable to denature
- c. Plasmids always integrate into the chromosomes, allowing scientists to inject them into predetermined cells
- d. Plasmids are able to adapt to any ploidy number

46. What is the difference between gene knock-out and knock-down? [1]

- a. Gene knock-out results in a temporary reduction of expression, and knock-down results in permanent inactivation of the gene
- b. Gene knock-out refers to changes made at the mRNA level, while knock-down refers to changes made at the DNA level
- c. Gene knock-out results in permanent inactivation of the gene, and knock-down results in temporary reduction of expression
- d. Gene knock-out removes the gene from the genome, and knock-down moves the gene downstream

47. When is a genetic knock-down more appropriate than knock-out? [1]

- a. When the gene of interest is conditionally expressed
- b. When the gene of interest is lethal if not present
- c. When the gene of interest is long in nucleotide numbers
- d. When the gene of interest is not being transcribed

2025 State Tournament - Div C -Georgia

Designer Genes - FRQ

**GEORGIA
SCIENCE
OLYMPIAD**



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Free response questions

1. The three nonsense codons in the genetic code are UAG, UAA, and UGA. Suppose a chemical mutagen causes frequent G to A mutations during DNA replication. Predict which stop codons would be expected to arise most often from these

mutations and explain why. In your response, compare the number of base changes required to produce each stop codon. [3 points]

In a certain species of flowers, pink flowers are dominant over white flowers and tall plants are dominant over dwarf plants. A tall pink flowered plant is crossed with a white flowered dwarf plant and the following progeny is obtained:

Pink, tall	131
Pink, short	47
White, tall	40
White, short	6

2. Assume that “A” and “a” correspond to dominant allele and recessive alleles for color, respectively. Assume that “B” and “b” correspond to dominant and recessive alleles for height, respectively. What are the most likely genotypes of the parents and progeny? **BONUS:** Test your genetic hypothesis with a chi-square test at the 0.05 significance level using the following chi-square table: [5 points]

Table 3.5 Critical values of the χ^2 distribution

df	P								
	0.995	0.975	0.9	0.5	0.1	0.05*	0.025	0.01	0.005
1	0.000	0.000	0.016	0.455	2.706	3.841	5.024	6.635	7.879
2	0.010	0.051	0.211	1.386	4.605	5.991	7.378	9.210	10.597
3	0.072	0.216	0.584	2.366	6.251	7.815	9.348	11.345	12.838
4	0.207	0.484	1.064	3.357	7.779	9.488	11.143	13.277	14.860
5	0.412	0.831	1.610	4.351	9.236	11.070	12.832	15.086	16.750
6	0.676	1.237	2.204	5.348	10.645	12.592	14.449	16.812	18.548
7	0.989	1.690	2.833	6.346	12.017	14.067	16.013	18.475	20.278
8	1.344	2.180	3.490	7.344	13.362	15.507	17.535	20.090	21.955
9	1.735	2.700	4.168	8.343	14.684	16.919	19.023	21.666	23.589
10	2.156	3.247	4.865	9.342	15.987	18.307	20.483	23.209	25.188
11	2.603	3.816	5.578	10.341	17.275	19.675	21.920	24.725	26.757
12	3.074	4.404	6.304	11.340	18.549	21.026	23.337	26.217	28.300
13	3.565	5.009	7.042	12.340	19.812	22.362	24.736	27.688	29.819
14	4.075	5.629	7.790	13.339	21.064	23.685	26.119	29.141	31.319
15	4.601	6.262	8.547	14.339	22.307	24.996	27.488	30.578	32.801

P, probability; df, degrees of freedom.

*Most scientists assume that, when $P < 0.05$, a significant difference exists between the observed and the expected values in a chi-square test.

ANSWER:

An experiment on corn kernels was conducted by crossing a pure-bred red-kernel strain (AABBCC) and a pure-bred white-kernel (aabbcc) strain. The F1 generation showed a phenotype of pink. The F2 generation produced a range of colors with 2 out of 128 offspring having a white phenotype identical to the parent.

- Based on the provided data, calculate the number of genes that contribute to the white kernel color in this population. [1 point]
- Based on the description which graph most nearly applies to the F2 generation of this population. [*1 point]

a.

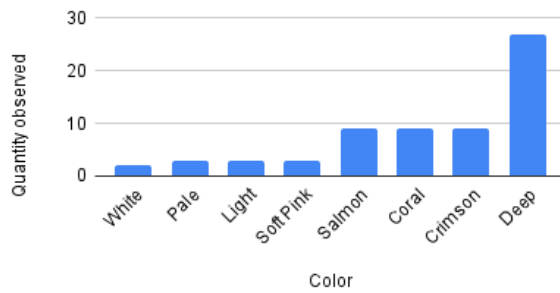
b.

Quantitv observed vs. Color

Quantitv observed vs. Color

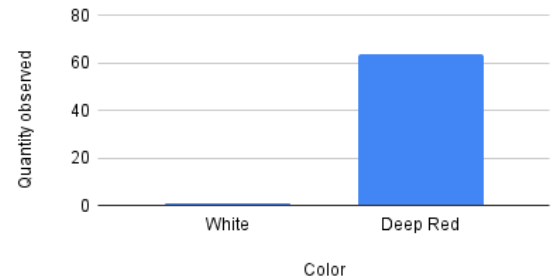
c.

Quantity observed vs. Color



d.

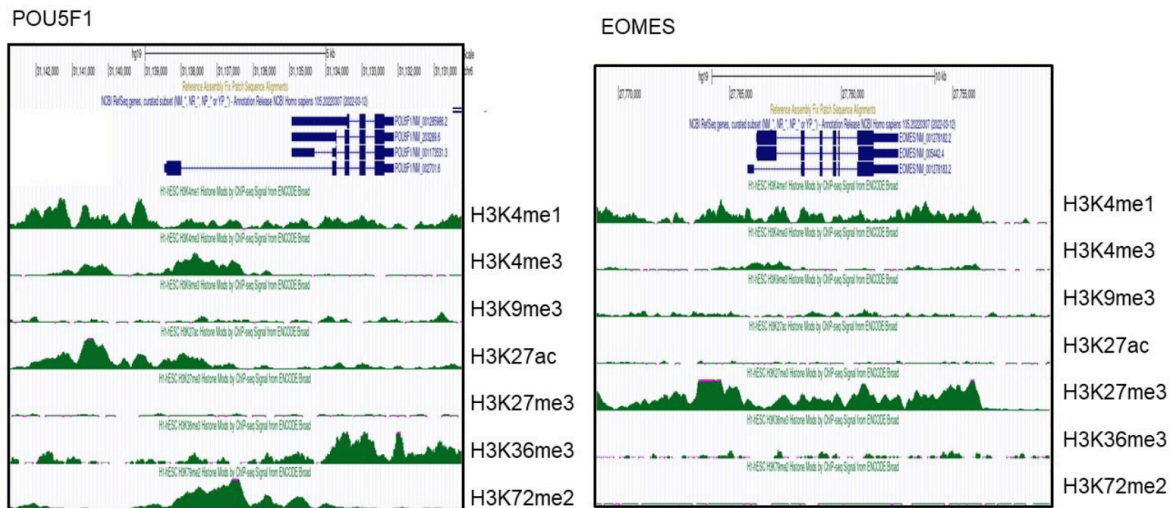
Quantity observed vs. Color



An engineer plans to build a border wall exactly in the middle of a grassland where a population of rabbits is inhabiting, thereby introducing a geographical barrier. A few years later, a highly contagious disease spreads and targets rabbits with a specific allele for fur color. Rabbits that have the fur color allele have a slight advantage regarding their metabolic function, a trait that is completely unrelated to fur color.

5. Mention at least two phenomena (evolutionary mechanisms) that happen to the population of rabbits in this situation [4 points]?

6. Shown below are histone modification profiles of genomic regions containing two genes of interest - POU5F1 and EOMES – in human embryonic stem cells. Which of these genes will be expressed in these cells? Explain. [*2 points]



7. During a PCR, there are three temperature dependent changes during the three specific steps of the process. Please name the step and why each step is temperature dependent (6 points).

8. Provide a brief explanation to the purpose of each given reagent for a standard PCR (5 points).

9. Briefly explain how Sanger sequencing works to label the DNA sequence (4 points).